

Polymer-Tethered Ligand–Receptor Interactions Between Surfaces

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ABSTRACT: We study the interaction between two parallel surfaces having a polymer-tethered ligand on one and a random distribution of receptors on the other. We examine the interplay between the specific ligand–receptor binding and the conformation degrees of freedom of the polymer tether, and address the difference between the cases of mobile (annealed) and immobile (quenched) receptors. The annealed case is solved exactly and the quenched case is treated by both Monte Carlo direct sampling and an analytical density expansion. The combination of the entropic repulsion due to chain confinement at small surface separations, and the attraction due to ligand–receptor binding which can take place at significant chain stretching, results in a minimum in the interaction free energy. For the same set of parameters, stronger binding is obtained for the annealed case than for the quenched case, reflecting the ability of the mobile receptors to migrate into the region of the ligand. In the quenched case, binding is limited by the availability of receptors within the reach of the ligand; for a given receptor density and binding energy, there exists an optimal chain length that yields the lowest minimum in the interaction free energy for the quenched case. © 2006 Wiley Periodicals, Inc. *J Polym Sci Part B: Polym Phys* 44: 2621–2637, 2006

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INTRODUCTION

Ligand–receptor binding is a fundamental problem in biology and underlies such important processes as cell adhesion and motility, cancer growth, and immune responses.^{1–3} Using micropipet aspiration,⁴ atomic force microscopy,^{5–7} surface forces apparatus,^{8–12} and optical and magnetic tweezer techniques,^{13,14} several groups have been able to measure the molecular forces between

proteins and ligands. Computational studies using molecular dynamics^{15,16} and stochastic simulations^{17–19} have provided more direct physical pictures of the binding and unbinding processes.

The direct interaction between a ligand and a receptor is typically specific, i.e., it acts only between specific pairs of molecules or residues on molecules, and is short-ranged. Such short-ranged, specific interactions can be modulated by attaching flexible tethers to the ligand. Tethered ligands and receptors are naturally present in biological systems,¹ e.g., on the surfaces of T lymphocyte cells and leukocyte cells.² They are also widely used and studied in engineered biomimetic systems, such as artificial amphiphilic bilayers,^{20,21} engineered extracellular matrix proteins,²² and polymersomes.^{23,24}

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Schneider et al.²⁰ used the surface forces apparatus to measure the interactions between hydrogen bonded, functionalized lipid bilayers. They found that upon separating the hydrogen bonded bilayers, some of the amphiphiles were actually pulled out of the membrane rather than having their hydrogen bonds broken. This happens in spite of the fact that the hydrophobic anchorage of the amphiphiles is energetically stronger than the hydrogen bonding between the head groups. The explanation of this somewhat surprising result undoubtedly lies in the details of the detachment mechanism, specifically in the interplay between the ability of the amphiphiles to be drawn from the membrane one methylene group at a time and the concerted nature of the hydrogen bonding of the terminal groups.

Using surface forces apparatus, Israelachvili and coworkers^{11,12} made the first direct measurement of the interaction potential between polymer-tethered ligands (biotin) on one surface and receptor sites (streptavidin) on an opposing surface. The flexible tether connecting the ligand allows it to reach receptor sites that are some distance away from the anchoring location of the tether on the surface, thus dramatically increasing the range of attraction between the two surfaces. These authors found that binding occurs at separations that are comparable to the fully stretched chain length of the tether polymer. The conformation degrees of freedom of the polymer tether were found to contribute significantly to the interaction potential between the two surfaces. The dynamics of how the polymer-tethered ligand pairs with a receptor was also studied by combining Monte Carlo simulation with reaction rate theory.¹² Tethered ligand–receptor binding was also used by Hammer and coworkers to engineer the adhesion between functionalized polymersome vesicles.^{24,25} These authors showed that the tether chain stretching contributes a significant part to the critical tensile force that is necessary to separate the polymersome vesicles.

The interaction between two surfaces due to tethered ligand–receptor binding involves two qualitatively different levels of physics. The specific binding between the ligand and the receptor is dictated by the chemical details of the ligand and receptor groups; the range of this interaction is a few Angstroms. On the other hand, the ability for the ligand and receptor to get close enough to within the interacting range depends on the length and flexibility of the tether. It also depends crucially on whether the receptor sites

are mobile or frozen on the substrate. Mobile receptor sites, for example, those attached to a fluid lipid membrane, will be attracted by the ligands, leading to a rearrangement of their distribution, as is taken advantage of in the design of some molecular imprinting systems.²⁶ For immobile receptor sites—either because they are positionally frozen on the substrate, as in the case of crosslinked polymersomes,²⁴ or because their diffusion is sufficiently slow when compared with the time scales for binding²⁷ and conformation relaxation of the tether—the tethered ligand has to reach for a receptor. The resulting interaction between the two surfaces reflects an interesting interplay between the microscopic, specific molecular binding, and the mesoscopic, generic behavior of the flexible tether: the polymer tether enables the ligand to bind with a receptor within a much larger range than the molecular binding range and mediates the interaction through chain stretching and confinement; at the same time, specific binding also alters the conformation of the polymer tether.

Despite its close connection to polymer statistics, previous studies on the problems involving tethered binding, for example, cell adhesion, treated the effects of chain flexibility as marginal or irrelevant.²⁸ Only recently have the effects of the tether chain been addressed in detail. Longo and Szleifer²⁹ studied the binding between tethered polymer layers and receptors in solution. These authors confirmed the result that binding is enhanced by the tether chain and found a nonmonotonic dependence in the number of bound receptors per unit area on the surface coverage of ligands. The optimal surface coverage for receptor binding is attributed to changes in chain configurations due to binding and interactions between the tethers. Chen and Dormidontova³⁰ studied the interactions between surfaces functionalized with tethered ligands and receptors. They examined the binding of mono- and multi-valent ligands, and specifically studied the difference between surfaces fully covered by receptors and those carrying fixed number of immobile, regularly patterned receptors. For either mono- or multi-valent ligands, the sparsely distributed receptor case is qualitatively different from the fully covered case, resulting in different binding fraction dependences on surface separation.

The combination of specific binding (a microscopic event) and conformation flexibility of tether molecules (a mesoscopic property) lead to different physical behavior than is characteristic of either

attribute alone. Since this combination is a recurring theme in a number of interesting systems, in this article we study a simple model that has both attributes to elucidate certain essential features in the adhesion behavior due to the interplay between the specific interaction and the conformation degrees of freedom. While our ultimate goal is to understand and predict the adhesion behavior in cells and biomimetic systems, real systems often involve complications because of a multitude of factors, such as interactions between the tethers, electrostatic effects, solvent structure effects, and fluctuation of the membrane surfaces. Therefore, we choose to examine an idealized model of surface interactions involving tethered ligand–receptor binding—a lattice model of a ligand at the end of a single flexible polymer attached to a surface interacting with a surface having a random distribution of receptor sites. Here, the configuration degrees of freedom include both the conformation of the tether and the distribution of the receptor sites. The model is shown schematically in Figure 1 and will be specified in more detail in the next section.

The article is organized as follows. In “Model and Theory”, we develop the theories for tethered ligand–receptor binding for both mobile and immobile receptors. The case of mobile receptor sites is solved exactly, while the case of immobile receptor sites is treated by a low density expansion and Monte Carlo sampling of the receptor distributions. In “Results and Discussion”, we present results for a few properties of interest, such as the interaction free energy between the surfaces, the fraction of bridges, as well as the effects of binding on the receptor distribution in the case of mobile receptors and the tether end distribution in the case of immobile receptors. Through discussions of these results, we wish to elucidate the effects of flexible tether on the ligand–receptor binding and to illustrate the key differences between systems with fixed versus mobile receptor sites. The article ends with a summary of the main results and a few concluding remarks.

MODEL AND THEORY

Model Description

Two parallel surfaces opposite to each other are separated by a distance h in units of the lattice spacing a . On the lower surface (at $z = 0$) is a single polymer tether consisting of N units, anchored

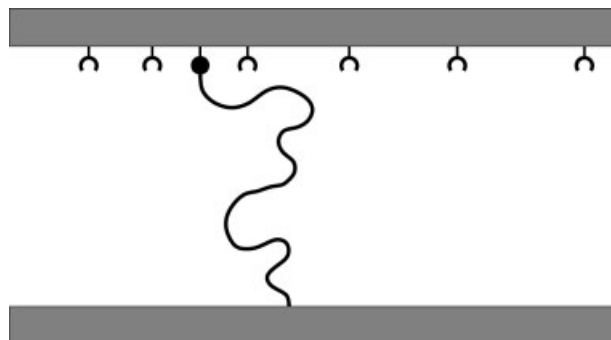


Figure 1. A schematic side view of the model.

to the surface at the origin of our coordinate system. At the end of this polymer is a single ligand, which can bind reversibly to a receptor on the upper surface, with a binding energy ϵ . The receptors on the upper surface are treated as a (otherwise) noninteracting two-dimensional lattice gas with excluded area. The spatial distribution of the receptors is conveniently represented using the occupation numbers $\{\sigma(\mathbf{r})\}$ [$\sigma(\mathbf{r}) = 1$ if site $\mathbf{r}$ is occupied by a receptor and 0 otherwise], where the bold-faced $\mathbf{r}$ denotes the two-dimensional coordinates (in units of the lattice spacing a) parallel to the receptor surface. We characterize the density of the receptors by their area fraction ϕ_0 , which is related to their area density ρ by $\phi_0 = \rho a^2$; henceforth, we will refer to ϕ_0 as the density of receptors. The interaction part of the Hamiltonian can be written as

$$H_{\text{int}}(\mathbf{r}_L, z) = -\epsilon \sum_{\mathbf{r}} \sigma(\mathbf{r}) \delta(\mathbf{r}_L - \mathbf{r}) \delta(z - h), \quad (1)$$

where $(\mathbf{r}_L, z)$ is the position of the ligand at the end of the tether chain.

There are two considerations in choosing a lattice model in this study. First, putting a polymer on a lattice facilitates the enumeration of polymer configurations and at the same time ensures that the finite extensibility of the tether is accounted for correctly. This latter aspect is an advantage over an equally convenient (in terms of mathematical tractability) Gaussian continuum model, which allows the chain end to be stretched to an arbitrary distance, albeit with an entropy penalty that increases with the distance. The finite extensibility becomes especially important when the tether is not very long and binding between the ligand and receptors is strong; in these cases, chain stretching is significant and cannot be captured by the Gaussian model.

Second, the lattice model reduces the number of parameters that need to be introduced. The size of the monomer units of the tether, the size of the ligand end group, and the size of the receptors are all taken to be the size of one lattice site. The range associated with the ligand–receptor interaction is taken to be one lattice spacing, so that if the tether chain terminates at the location on the receptor surface that is occupied ($\sigma = 1$), then it is taken as being bound with energy ϵ , otherwise, it is taken as unbound. While including the different lengths associated with the sizes of the various moieties and interaction ranges would undoubtedly be a more realistic representation of the actual physical or biophysical systems, having a minimal set of parameters in a model allows us to focus on the essential physics of the problem.

Although our model describes the interaction between a single tethered ligand and the receptor surface, the results are applicable to systems where the ligand density is low and receptors greatly outnumber ligands; in such cases, the different ligands can be considered independent and their contributions to the total interaction between the two surfaces are additive. We note that, even without direct interactions among the ligands, when the ligand density is high, the competition between different ligands to bind with the same receptor and vice versa, introduces nontrivial correlation effects. We leave the treatment of the problem of multiple ligands and receptors to a later publication.

For a given distribution of receptors $\sigma(\mathbf{r})$, the partition function is given by

$$Q[\sigma(\mathbf{r}), h, \epsilon] = \sum_{\mathbf{r}} \sum_{0 \leq z < h} G(\mathbf{r}, z) + \sum_{\mathbf{r}} G(\mathbf{r}, h) e^{\beta \epsilon \sigma(\mathbf{r})}. \quad (2)$$

Here, $G(\mathbf{r}, z)$ is the Green function of the polymer chain; the first term gives the statistical weight of chain conformations where the ligand (at the chain end) is located anywhere between the two surfaces ($0 \leq z < h$); the second term gives the statistical weight that the ligand reaches the receptor surface ($z = h$). From the definition of $\sigma(\mathbf{r})$, we have

$$e^{\beta \epsilon \sigma(\mathbf{r})} = 1 + \sigma(\mathbf{r})(e^{\beta \epsilon} - 1),$$

hence

$$Q[\sigma, h, \epsilon] = Q_0 \left[1 + (e^{\beta \epsilon} - 1) \sum_{\mathbf{r}} P_0(\mathbf{r}) \sigma(\mathbf{r}) \right]. \quad (3)$$

In the equation above,

$$Q_0 = \sum_{\mathbf{r}} \sum_{0 \leq z \leq h} G(\mathbf{r}, z)$$

is the partition function of the polymer chain in the absence of ligand–receptor binding, and

$$P_0(\mathbf{r}) = G(\mathbf{r}, h)/Q_0$$

is the probability of finding the free end at the lateral position $\mathbf{r}$ on the receptor surface in the absence of binding, which describes the end distribution of the polymer tether on the receptor surface. For an ideal random walk chain on a lattice, Q_0 and $P_0(\mathbf{r})$ can be calculated analytically; we provide the calculation in Appendix A.

The relevant free energy for the interaction between the two surfaces is the free energy at a separation h relative to that at infinite separation. Depending on whether the receptors are mobile or frozen, we have two scenarios henceforth termed “annealed” and “quenched”. In the annealed case, the average over the receptor distribution is performed on the partition function and the free energy is

$$F(h, \epsilon) = -kT \left\{ \ln \langle Q[\sigma(\mathbf{r}), h, \epsilon] \rangle_{\{\sigma\}} - \ln \langle Q[\sigma(\mathbf{r}), h = \infty, \epsilon] \rangle_{\{\sigma\}} \right\}. \quad (4)$$

In the quenched case, we need to average over the free energy for each receptor distribution; thus the free energy is

$$F(h, \epsilon) = -kT \left\{ \langle \ln Q[\sigma(\mathbf{r}), h, \epsilon] \rangle_{\{\sigma\}} - \langle \ln Q[\sigma(\mathbf{r}), h = \infty, \epsilon] \rangle_{\{\sigma\}} \right\}. \quad (5)$$

Of special interest is the probability for forming a bridge connecting the two surfaces due to ligand–receptor binding. This is given by

$$f = \left\langle \sum_{\mathbf{r}} \sigma(\mathbf{r}) \delta(\mathbf{r} - \mathbf{r}_L) \delta(z - h) \right\rangle \quad (6)$$

In a system with multiple but independent tethered ligands, this probability is just the fraction of chains that form bridges due to ligand–receptor

binding. For this reason, we term this quantity the binding fraction in subsequent discussions. Since f is conjugate to the binding energy (cf. eq 1), we have

$$f = -\frac{\partial F}{\partial \epsilon} \quad (7)$$

for both annealed and quenched cases.

We note that, while the quenched case is realized when both receptors and ligands are immobile, the annealed case corresponds to systems where either ligands or receptors are mobile. Therefore, our treatment of mobile receptors also applies to the case of immobile receptors with mobile ligands.⁴⁰

Next we discuss the different averages over the receptor distribution.

Mobile Receptors

For mobile receptors, the average of the chain partition function in eq 4 (defined in eqs 2 or 3) can be easily performed. However, it is worth noting that the proper average is a grand canonical average, as the area of the receptor surface where ligand-receptor binding takes place is logically viewed as a small part connected to a large reservoir of receptors having a fixed density ϕ_0 . The proper statistical average of a property X is therefore

$$\langle X \rangle_{\{\sigma\}} = \frac{1}{\Xi_0} \sum_{\{\sigma\}} X(\sigma) \exp \left[\beta \mu \sum_{\mathbf{r}} \sigma(\mathbf{r}) \right], \quad (8)$$

where

$$\Xi_0 = \sum_{\{\sigma\}} \exp \left[\beta \mu \sum_{\mathbf{r}} \sigma(\mathbf{r}) \right] = [1 + \exp(\beta \mu)]^M. \quad (9)$$

is the grand partition function of the receptors in an area with M sites, and the chemical potential is related to the average density ϕ_0 from lattice gas statistics,

$$\beta \mu = \ln \left(\frac{\phi_0}{1 - \phi_0} \right). \quad (10)$$

Using eq 3 to perform the average in eq 4 defined by eq 8, we obtain

$$\begin{aligned} F(h, \epsilon) &= -kT \ln \frac{Q_0(h)}{Q_0(\infty)} \\ &\quad - kT \ln \left[1 + (e^{\beta \epsilon} - 1) \sum_{\mathbf{r}} \frac{e^{\beta \mu}}{1 + e^{\beta \mu}} P_0(\mathbf{r}) \right] \\ &= -kT \ln \frac{Q_0(h)}{Q_0(\infty)} \\ &\quad - kT \ln \left[1 + \phi_0 (e^{\beta \epsilon} - 1) \frac{Q_1}{Q_0} \right] \\ &\equiv F_0(h) + F_b(h, \epsilon), \end{aligned} \quad (11)$$

where for convenience we have defined

$$Q_1 = \sum_{\mathbf{r}} G(\mathbf{r}, h), \quad (12)$$

which is the statistical weight of chain configurations with the ligand located anywhere on the receptor surface in the absence of binding. The first term in eq 11 is the free energy of interaction between a bare surface and a surface with a polymer attached to it; this term is independent of the ligand-receptor binding. The second term is the excess free energy due to ligand-receptor binding; we refer to this contribution as the binding free energy.

The binding fraction f is obtained from eqs 7 and 11

$$f = -\frac{\partial F_b}{\partial \epsilon} = \frac{\phi_0 e^{\beta \epsilon} Q_1 / Q_0}{1 + \phi_0 (e^{\beta \epsilon} - 1) Q_1 / Q_0}. \quad (13)$$

Because the receptors are mobile, the local receptor density in the area directly above the tethered ligand is enhanced over the reservoir density because of the attraction by the ligand. To capture this effect, we calculate the local receptor density on the receptor surface. This is conveniently done by introducing a spatially dependent chemical potential $\mu'(\mathbf{r})$ into the binding part of the free energy (cf. eq 11) so that

$$\begin{aligned} F_b[h, \epsilon, \mu'(\mathbf{r})] \\ = -kT \ln \left[1 + (e^{\beta \epsilon} - 1) \sum_{\mathbf{r}} \frac{e^{\beta \mu + \beta \mu'(\mathbf{r})}}{1 + e^{\beta \mu + \beta \mu'(\mathbf{r})}} P_0(\mathbf{r}) \right]. \end{aligned}$$

Since $\mu + \mu'(\mathbf{r})$ is conjugate to $\phi(\mathbf{r})$, the receptor density enhancement is given by

$$\begin{aligned}\phi(\mathbf{r}) - \phi_0 &= - \left. \frac{\partial F}{\partial \mu'(\mathbf{r})} \right|_{\mu'(\mathbf{r})=0} \\ &= \frac{\phi_0(1 - \phi_0)P_0(\mathbf{r})(e^{\beta\epsilon} - 1)}{1 + \phi_0(e^{\beta\epsilon} - 1)Q_1/Q_0}.\end{aligned}\quad (14)$$

$\Delta\phi(\mathbf{r}) = \phi(\mathbf{r}) - \phi_0$ represents the enhancement of local receptor density in the vicinity of the ligand. Since there is only a single ligand, $\Delta\phi(\mathbf{r})$ is simply due to the receptor bound by the ligand.

Immobile Receptors

In this subsection we consider the case of immobile receptors. Here, the key physical quantities of interest are the average free energy over the configuration of receptors $\sigma(\mathbf{r})$ given in eq 5,

$$\begin{aligned}F(h, \epsilon) &= -kT \ln \frac{Q_0(h)}{Q_0(\infty)} \\ &\quad - kT \left\langle \ln \left[1 + (e^{\beta\epsilon} - 1) \sum_{\mathbf{r}} \sigma(\mathbf{r})P_0(\mathbf{r}) \right] \right\rangle_{\{\sigma\}} \\ &\equiv F_0(h) + F_b(h, \epsilon)\end{aligned}\quad (15)$$

and the average binding fraction is

$$f = - \frac{\partial F(h, \epsilon)}{\partial \epsilon} = \left\langle \frac{e^{\beta\epsilon} \sum_{\mathbf{r}} P_0(\mathbf{r}) \sigma(\mathbf{r})}{1 + \sum_{\mathbf{r}} P_0(\mathbf{r}) \sigma(\mathbf{r})(e^{\beta\epsilon} - 1)} \right\rangle_{\{\sigma\}} \quad (16)$$

As in the case of mobile receptors, the first term in the free energy is the free energy of interaction between the two surfaces due to the polymer tether in the absence of ligand–receptor binding. The second term is the excess free energy due to polymer-mediated ligand–receptor binding.

In considering the quenched average, we treat each realization of the receptor configuration to be that of a noninteracting lattice gas. In other words, each quenched configuration can be viewed as generated by freezing the corresponding configuration in a lattice gas with average density ϕ_0 produced by a chemical potential μ in a grand canonical ensemble. Therefore, the same average as defined in eq 8 applies here.

Because of the complex dependence of the expressions, averaging over the configuration of the receptors cannot be performed analytically, hence the above equations cannot be written in closed forms. Here, we employ two simple methods to evaluate the quenched average. The first method, Monte Carlo sampling, is to generate a large nonexhaustive set of random receptor configurations according to the binomial distribution as is appropriate for the lattice-gas statistics with excluded area. This direct sample averaging yields fairly accurate results for a reasonably large number of samples. For example, 10,000 configurations gives free energy values that are reproducible to within 0.1% for receptor densities greater than 0.001; the numerical accuracy improves at higher receptor densities.

When the receptor density is low, we can develop a density expansion in terms of

$$\phi_0 = \frac{e^{\beta\mu}}{1 + e^{\beta\mu}}.$$

As this method may have applications to a wide class of problems involving quenched disorder, we briefly describe it in the calculation of the second term of the free energy F_b in eq 15.

The key step in developing the density expansion is to rewrite the summation in eq 8 over configurations $\{\sigma\}$ as summations over the positions of the receptors and their total number within the area, which gives

$$\begin{aligned}\langle X(\sigma) \rangle &= \sum_{0 \leq n \leq M} \frac{1}{n!} \frac{e^{\beta\mu n}}{(1 + e^{\beta\mu})^M} \sum_{\mathbf{r}_1} \sum_{\mathbf{r}_2 \neq \mathbf{r}_1} \cdots \sum_{\mathbf{r}_n \neq \mathbf{r}_1, \dots, \mathbf{r}_{n-1}} X(\sigma) \\ &= \sum_{0 \leq n \leq M} \frac{1}{n!} \left(\frac{\phi_0}{1 - \phi_0} \right)^n (1 - \phi_0)^M \sum_{\mathbf{r}_1} \sum_{\mathbf{r}_2 \neq \mathbf{r}_1} \cdots \sum_{\mathbf{r}_n \neq \mathbf{r}_1, \dots, \mathbf{r}_{n-1}} X(\sigma) \\ &= \sum_{0 \leq n \leq M} \sum_{k=0}^{M-n} \frac{(-1)^k (M-n)!}{n! k! (M-n-k)!} \phi_0^{n+k} \sum_{\mathbf{r}_1} \sum_{\mathbf{r}_2 \neq \mathbf{r}_1} \cdots \sum_{\mathbf{r}_n \neq \mathbf{r}_1, \dots, \mathbf{r}_{n-1}} X(\sigma).\end{aligned}\quad (17)$$

Next, note that the occupation number can be written as

$$\sigma(\mathbf{r}) = \sum_{i=1}^n \delta(\mathbf{r} - \mathbf{r}_i), \quad (18)$$

where $\mathbf{r}_i$ is the position of the i th receptor, the average of the logarithm in eq 15 can be expanded as

$$\begin{aligned} & \left\langle \ln \left[1 + \sum_{\mathbf{r}} P_0(\mathbf{r}) \sigma(\mathbf{r}) (e^{\beta\epsilon} - 1) \right] \right\rangle_{\{\sigma\}} \\ &= \sum_{0 \leq n \leq M} \sum_{k=0}^{M-n} \frac{(-1)^k (M-n)!}{n! k! (M-n-k)!} \phi_0^{n+k} \\ & \sum_{\mathbf{r}_1} \sum_{\mathbf{r}_2 \neq \mathbf{r}_1} \cdots \sum_{\mathbf{r}_n \neq \mathbf{r}_1, \dots, \mathbf{r}_{n-1}} \ln \left[1 + \sum_{i=1}^n P_0(\mathbf{r}_i) (e^{\beta\epsilon} - 1) \right]. \end{aligned} \quad (19)$$

Collecting the terms according to the powers of ϕ_0 , we obtain the binding free energy

$$\begin{aligned} F_b(h, \epsilon) &= -kT \left\langle \ln \left[1 + \sum_{\mathbf{r}} P_0(\mathbf{r}) \sigma(\mathbf{r}) (e^{\beta\epsilon} - 1) \right] \right\rangle_{\{\sigma\}} \\ &= -kT \sum_{n=1}^{\infty} B_n (\beta\epsilon, h) \phi_0^n. \end{aligned} \quad (20)$$

The coefficients B_n are evaluated in Appendix B. The first two terms are:

$$\begin{aligned} B_1 &= \sum_{\mathbf{r}} \ln [1 + P_0(\mathbf{r}) (e^{\beta\epsilon} - 1)], \quad (21) \\ B_2 &= -\frac{1}{2} \\ & \times \left\{ \sum_{\mathbf{r}_1} \sum_{\mathbf{r}_2} \ln \frac{[1 + P_0(\mathbf{r}_1) (e^{\beta\epsilon} - 1)][1 + P_0(\mathbf{r}_2) (e^{\beta\epsilon} - 1)]}{1 + [P_0(\mathbf{r}_1) + P_0(\mathbf{r}_2)] (e^{\beta\epsilon} - 1)} \right. \\ & \quad \left. - \sum_{\mathbf{r}} \ln \frac{[1 + P_0(\mathbf{r}) (e^{\beta\epsilon} - 1)]^2}{1 + 2P_0(\mathbf{r}) (e^{\beta\epsilon} - 1)} \right\}. \end{aligned} \quad (22)$$

The binding fraction can be calculated similarly from eq 16 to give

$$f = \sum_{n=1}^{\infty} C_n (\beta\epsilon, h) \phi_0^n. \quad (23)$$

The first two coefficients are

$$C_1 = \sum_{\mathbf{r}} \frac{P_0(\mathbf{r}) e^{\beta\epsilon}}{1 + P_0(\mathbf{r}) (e^{\beta\epsilon} - 1)}, \quad (24)$$

$$\begin{aligned} C_2 &= \sum_{\mathbf{r}} \left[\frac{P_0(\mathbf{r}) e^{\beta\epsilon}}{1 + P_0(\mathbf{r}) (e^{\beta\epsilon} - 1)} - \frac{P_0(\mathbf{r}) e^{\beta\epsilon}}{1 + 2P_0(\mathbf{r}) (e^{\beta\epsilon} - 1)} \right] \\ &+ \frac{1}{2} \sum_{\mathbf{r}_1} \sum_{\mathbf{r}_2} \left[\frac{[P_0(\mathbf{r}_1) + P_0(\mathbf{r}_2)] e^{\beta\epsilon}}{1 + [P_0(\mathbf{r}_1) + P_0(\mathbf{r}_2)] (e^{\beta\epsilon} - 1)} \right. \\ & \quad \left. - \frac{P_0(\mathbf{r}_1) e^{\beta\epsilon}}{1 + P_0(\mathbf{r}_1) (e^{\beta\epsilon} - 1)} - \frac{P_0(\mathbf{r}_2) e^{\beta\epsilon}}{1 + P_0(\mathbf{r}_2) (e^{\beta\epsilon} - 1)} \right]. \end{aligned} \quad (25)$$

It is informative to compare the end position distribution of the polymer tether between the quenched and the annealed cases. In the annealed case, once the end of the polymer reaches the receptor surface, its lateral distribution is unaffected by binding with the mobile receptors, as the receptors can move along with the ligand. In the quenched case, however, the chain may need to stretch in search for a receptor to bind. This effect leads to an altered end distribution; we now obtain an expression for this probability distribution. We rewrite the partition function eq 2 as

$$Q = Q_0 - Q_1 + Q_0 \sum_{\mathbf{r}} P_0(\mathbf{r}) [1 + \sigma(\mathbf{r}) (e^{\beta\epsilon} - 1)]. \quad (26)$$

Here, $Q_0 - Q_1$ is the statistical weight of the chain configurations with the ligand not reaching the receptor surface ($0 \leq z < h$); the term in the summation is the statistical weight of configurations with the ligand reaching $\mathbf{r}$ on the receptor surface. For a given receptor configuration, the probability that the ligand reaches $\mathbf{r}$ on the receptor surface is therefore

$$P_{\sigma}(\mathbf{r}) = \frac{Q_0 P_0(\mathbf{r}) [1 + (e^{\beta\epsilon} - 1) \sigma(\mathbf{r})]}{Q_0 - Q_1 + Q_0 \sum_{\mathbf{r}'} P_0(\mathbf{r}') [1 + \sigma(\mathbf{r}') (e^{\beta\epsilon} - 1)]} \quad (27)$$

The quench-averaged probability is thus

$$\begin{aligned} P(\mathbf{r}) &= \left\langle \frac{Q_0 P_0(\mathbf{r}) [1 + (e^{\beta\epsilon} - 1) \sigma(\mathbf{r})]}{Q_0 - Q_1 + Q_0 \sum_{\mathbf{r}'} P_0(\mathbf{r}') [1 + \sigma(\mathbf{r}') (e^{\beta\epsilon} - 1)]} \right\rangle_{\{\sigma\}} \\ &= \left\langle \frac{P_0(\mathbf{r}) [1 + (e^{\beta\epsilon} - 1) \sigma(\mathbf{r})]}{1 + (e^{\beta\epsilon} - 1) \sum_{\mathbf{r}'} P_0(\mathbf{r}') \sigma(\mathbf{r}')} \right\rangle_{\{\sigma\}} \end{aligned} \quad (28)$$

A density expansion can be similarly developed for this probability; however, the results are a bit cumbersome. Therefore, we evaluate this quantity using the Monte Carlo sampling.

RESULTS AND DISCUSSION

In this section, we use the solutions developed in the previous section to examine a number of properties related to the interaction between two surfaces due to polymer-tethered ligand–receptor binding. We focus on the free energy of interaction between the surfaces and the binding fraction of the tethered ligand as a function of the separation h , molecular binding energy ϵ , the receptor density ϕ_0 , as well as the chain length N .

As discussed in the previous section, the interaction free energy (for both annealed and quenched cases) consists of two contributions, one simply due to the presence of a polymer chain, and the other due to the ligand–receptor binding mediated by the polymer (cf. eqs 11 and 15). The interaction due to the tether polymer is purely repulsive (here we assume no adsorption of the polymer to either surface) because of entropy loss when the polymer is confined between two surfaces; the repulsion becomes significant when the separation between the surfaces becomes less than the average end-to-end distance of the polymer. On the other hand, the contribution due to ligand–receptor binding is attractive. Binding is appreciable if the tethered ligand can reach a receptor on the other surface, and this occurs when the binding energy ϵ is sufficient to compensate for the increase in the stretching free energy. For moderate to large binding energies ($10kT$ to $30kT$) and relatively short chains ($N < 50$), the onset takes place at a distance that can be a significant fraction of the fully stretched length of the polymer. (If we took a Gaussian chain model, we would obtain the simple scaling $h \sim \sqrt{N\beta\epsilon}$, but at this distance the Gaussian model is no longer valid for large $\beta\epsilon$ and short chain length N .) As the separation decreases, the binding fraction increases, with an attendant decrease in the binding free energy. For large enough binding energy, there can be a considerable range of separation in which the binding free energy becomes appreciable but the confinement entropy for the chain is not yet significant. At sufficiently close separation, the entropic repulsion dominates and the overall interaction becomes repulsive. This competition between entropic repulsion and attractive binding leads to a free energy minimum.

The behavior of the total interaction free energy and the binding fraction as functions of the surface separation is shown in Figure 2 for two binding energies $\epsilon = 10kT$ and $\epsilon = 15kT$, for both annealed and quenched cases. The curves are

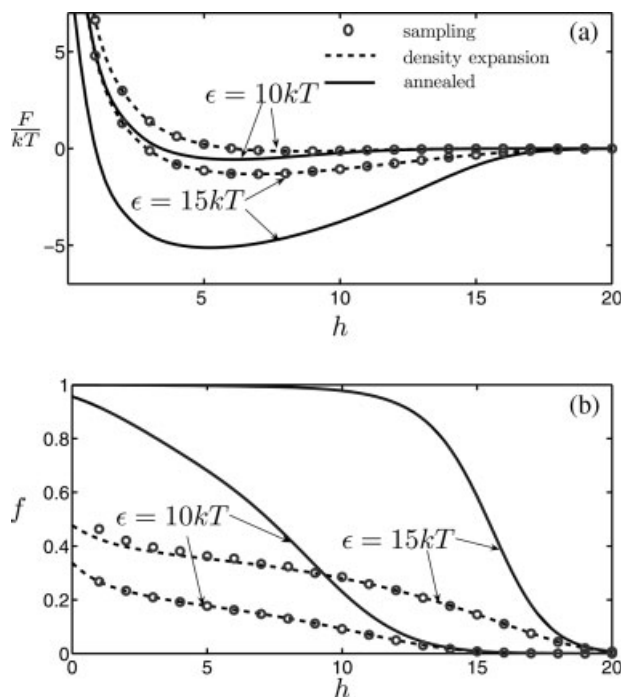


Figure 2. Dependence of the total interaction free energy and the binding fraction on the surface separation for two values of binding energy, $\epsilon = 10kT$ and $\epsilon = 15kT$. The tether length is 50 units, receptor density is 0.001. Results for the annealed case are given by the solid line. Direct sampling results for the quenched case are indicated by the circles and the dashed lines are results obtained from the two-term density expansion.

calculated using a tether length $N = 50$. We choose a low receptor density $\phi_0 = 0.001$ and moderate binding energies, as the density expansion is accurate under these conditions for the quenched cases, which is included for comparison.

From the figure it is clear that for both values of the binding energy, stronger binding is obtained for the annealed than for the quenched case. This effect reflects the ability of mobile receptors to migrate into the region of the tethered ligand, which only incurs a small entropic cost for configuration reorganization when the receptor density is not too low. For the quenched case, on the other hand, the chain has to stretch to find a receptor to bind at low receptor coverage ($\phi_0 = 0.001$ in this case), which is accompanied by a considerable reduction in the entropy of the polymer (the polymer is more stretched laterally). The separation at which the binding fraction starts to increase appreciably occurs at $h \approx 15$ for $\beta\epsilon = 10$ and $h \approx 20$ for $\beta\epsilon = 15$; the chain stretching is significant in both cases. Interestingly, these values

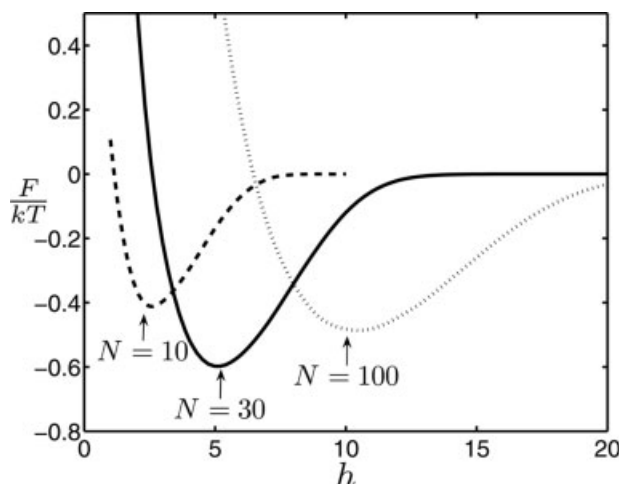


Figure 3. Effect of the tether length N on the free energy of interaction for the quenched case with $\phi_0 = 0.003$ and $\epsilon = 10kT$. Results are obtained using the two-term density expansion for tether lengths $N = 10$ (dashed line), $N = 30$ (solid line), and $N = 100$ (dotted line).

appear to be the same for the annealed case as for the quenched case.

A major difference is that in the annealed case, the binding fraction increases rapidly to near unity as the separation decreases, while in the quenched case, the increase is very gradual, and even at zero separation the binding fraction is considerably below unity. This lower saturation value for the quenched case reflects the limitation of binding by the availability of receptors within the distance that can be reached by the tethered ligand; the saturation value is set by the probability of finding at least one receptor within the reach of the ligand. For low receptor densities, this probability can be significantly less than one even at zero separation.

We also note that the difference between the annealed and the quenched cases is more pronounced at larger ϵ . The reason is that increasing the binding energy has a much more dramatic effect on binding in the annealed case than in the quenched case. This point is discussed in more detail following Figure 5.

Finally, we note that for the quenched case, results from the low density expansion using only two terms agree with those from direct Monte Carlo sampling calculation remarkably well. Only for the larger binding energy $\beta\epsilon = 15$ at very small h , do we observe some difference between the two calculations. The validity of the density expansion will be further discussed later.

In the quenched case, binding depends strongly on the number of receptors within the reach of the tethered ligand. This number increases as the surfaces come closer and/or as the tether length increases, implying lower binding free energy under these conditions. On the other hand, the entropy cost for confining a polymer chain becomes larger for longer chains and closer separations. Therefore, one can expect an “optimal” chain length, at which the balance between the free energy cost due to confinement and the gain due to binding leads to a lowest minimum. This effect is demonstrated in Figure 3, where the quenched free energy is calculated (using the two-term low density expansion) for three chain lengths: $N = 10, 30$, and 100 . The binding energy is $\epsilon = 10kT$, and the receptor density is $\phi_0 = 0.003$. Clearly, a lower free energy minimum is obtained for the $N = 30$ tether.

From the equilibrium free energy of interaction between the two surfaces, an interaction force can be obtained from

$$\tau = \frac{\partial F}{\partial h},$$

where we use the sign convention that the tension is positive if the surfaces are attracting each other. In Figure 4, we show the force calculated for $\epsilon = 15kT$, $N = 30$, and $\phi_0 = 0.003$. Of particular

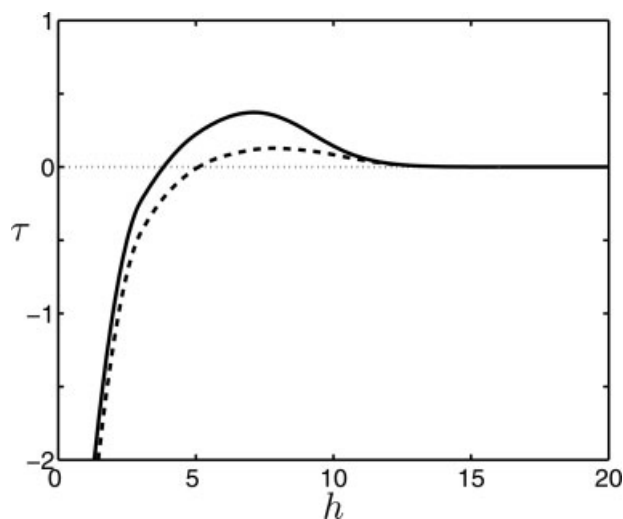


Figure 4. Equilibrium tension τ as a function of the separation h . τ is given in unit of kT per unit length a . The parameters are $N = 30$, $\phi_0 = 0.003$, and $\epsilon = 15kT$. The solid curve corresponds to the annealed case and the dashed curve corresponds to the quenched case. Results for the quenched case is obtained using the two-term density expansion.

interest is the critical tension for separating the two surfaces, which corresponds to the maximum in the force curve. Clearly, the use of a flexible polymer tether has a dramatic effect on the critical tension. While the molecular ligand–receptor bond is very short ranged and is quite strong, the ensemble averaged, polymer-mediated binding force is much weaker and has a much longer range.⁴¹ It is tempting to draw a comparison between the theoretical predictions and experiments. However, such a comparison is hampered by two serious complications in experiments. First, most experiments use strong ligand–receptor bonds, where binding is nearly irreversible and is only limited by diffusion. Second, under any finite tensile force, a ligand–receptor bond is only metastable and has a finite life time, which decreases as the force increases.³¹ When the force is shared by many bonds as is usually the case in experiments, the breaking of one bond shifts the load to the remaining bonds and increases the force per bond abruptly. This can lead to an avalanche effect as first discussed by Bell³² and more recently by Seifert.³³ Nevertheless, the result that the polymer tether has a significant effect on the critical tension is consistent with the conclusions reached in the experimental studies of Hammer and coworkers.²⁵ Furthermore, the equilibrium force separation curve provides a useful reference for addressing the various dynamic and nonequilibrium effects in experiments.

The receptor density ϕ_0 and molecular binding energy ϵ strongly influence the interaction between the surfaces. As briefly discussed earlier in this section, the total free energy of interaction consists of two parts, one due to the polymer tether in the absence of ligand–receptor binding, and the other due to polymer-mediated ligand–receptor binding, which we have termed the binding free energy F_b . The first contribution (the first term in eqs 11 and 15) is independent of binding; all the dependence on ϕ_0 and ϵ is contained in F_b . In the following discussions, we focus on the latter part.

Figure 5 shows the dependence of binding free energy F_b and binding fraction f on the binding energy ϵ . The tether length $N = 50$, receptor density $\phi_0 = 0.001$, and surface separation is fixed at $h = 10$. At smaller ϵ ($< 10kT$), the gain in binding energy is insufficient to compensate for the stretching energy; hence, little binding is observed in either the annealed or the quenched case. At $\epsilon \approx 10kT$, there is a sharp transition in the annealed case where f increases rapidly and saturates to nearly unity. In the quenched case, the

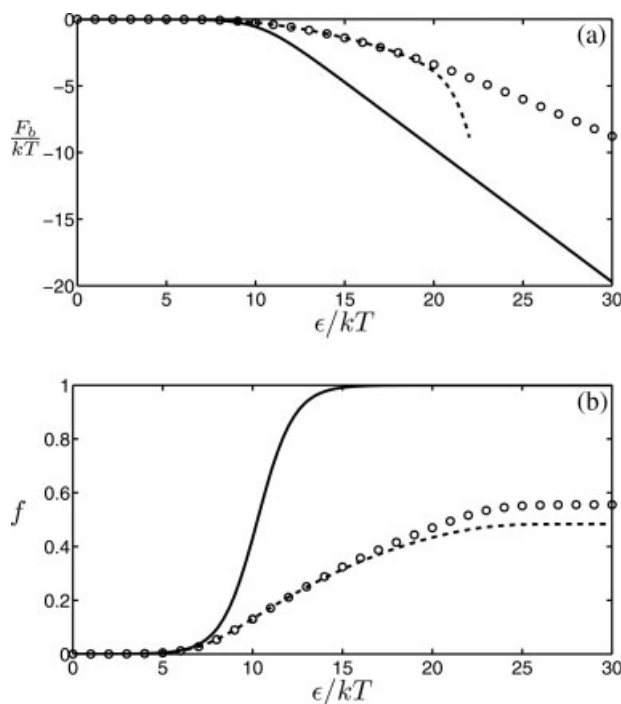


Figure 5. Dependence of the binding free energy F_b and the binding fraction f on the molecular binding energy ϵ . The other parameters are $N = 50$, $\phi_0 = 0.001$, and $h = 10$. The meaning of curves are the same as in Figure 2.

increase is less dramatic and the binding fraction saturates at approximately 0.5; this is consistent with the argument we made earlier that the maximum binding fraction in the quenched case is limited by the receptor density. No matter how large the binding energy is, binding will not saturate if there are not enough receptors within reach of the polymer tether. The restricted availability of receptors limits the effect of increased binding energy in the quenched case, when compared with the annealed case. Therefore, the difference between the quenched and the annealed case becomes amplified at larger ϵ , as observed in Figure 2.

An interesting feature concerning the binding free energy is that once significant binding occurs, the binding free energy decreases linearly for both annealed and quenched cases. In the annealed case, the slope is nearly one (with a negative sign), suggesting that any additional increase in the molecular binding energy is almost completely converted into the binding free energy. In the quenched case, on the other hand, the slope is significantly less than one, implying that a significant portion of the additional binding

energy—nearly one half for the set of parameters studied—goes into other forms of free energy, presumably stretching of the polymer tether. For large ϵ , the binding part of the free energy is approximately $F_b \approx -f\epsilon$. Thus, the difference in the slope between these two cases reflects primarily the difference in the binding fraction f . At a given ϕ_0 , f saturates at large ϵ (to unity in the annealed case and less than 1 for the quenched case), so the difference in the binding fraction saturates with increasing ϵ . However, the difference in the binding free energy increases because of the linear dependence on ϵ .

Included in Figure 5 for the quenched case are results from the second order low density expansion. We note that the expansion works quite well for lower binding energies but begins to fail at $\beta\epsilon \approx 20$. We can work out a self-consistent criterion for the validity of this approximation. Intuitively, since the low density expansion is carried out around the no binding limit, in order for the first couple of terms to capture the full behavior, we require that the binding fraction be small.

From Figures 2, 5, and 6 (to be discussed later), significant deviations from the low density expansion occurs at approximately $f \approx 0.4$. It can be shown that the first and second order terms in eq 23 are of opposite signs. Thus, at $f = 0.4$, the first order term is larger than this value. We can therefore empirically set the validity of the low density expansion to be when

$$C_1\phi_0 < 0.4, \quad (29)$$

where C_1 is the first order expansion coefficient given by eq 24. The above inequality provides a criterion for the validity of the low density expansion in terms of ϕ_0 , $\beta\epsilon$, N , and h (the latter two implicit in the probability $P_0(\mathbf{r})$ for the ligand to reach a lateral position $\mathbf{r}$ on the receptor surface). Unfortunately, since C_1 is not expressible in terms of a close-form function of N , $\beta\epsilon$, and h , we do not have a simple close-form relation. However, C_1 can be readily evaluated numerically, so in principle, given any three parameters, one can determine the value of the fourth parameter for a self-consistency check of the low density expansion.

The dependences of binding fraction f and binding free energy F_b on the receptor density ϕ_0 are illustrated in Figure 6 for $N = 50$, $h = 10$, and $\beta\epsilon = 10$. The meaning of the curves are the same as in Figure 5. As expected, the quenched and

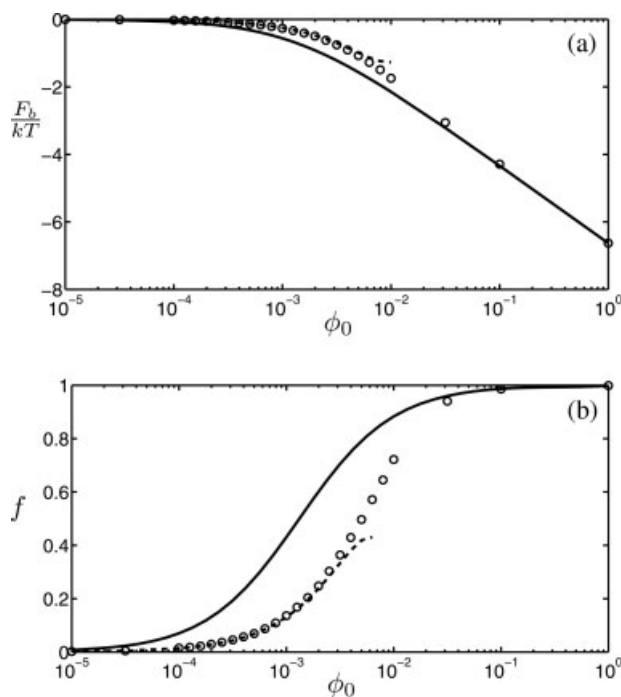


Figure 6. Dependence of the binding energy and the binding fraction on the receptor density ϕ_0 . The other parameters are $N = 50$, $\epsilon = 10kT$, and $h = 10$. The meaning of curves are the same as in Figure 2.

annealed cases become identical when the receptor surface is fully occupied. From the figure we see that the two become nearly indistinguishable at $\phi_0 \sim 0.1$. This is because at ϕ_0 higher than this value, there are a large number of receptors within the reach of the polymer.⁴² For intermediate receptor densities (10^{-4} to 10^{-1}), however, the difference between the two is evident, especially in the binding fraction. As receptors are attracted to the vicinity of the ligand in the annealed case, we see that there is significant binding even at very low receptor density $\phi_0 \approx 10^{-4}$ for the annealed case, while little binding takes place for the quenched case. At sufficiently high receptor density, the low density expansion breaks down as expected. For the current set of parameters, this occurs at $\phi_0 \approx 0.01$.

As alluded to earlier, in the annealed case, the tethered ligand can attract receptors to its vicinity, thereby increasing the probability of finding a receptor, while in the quenched case, polymer tether stretches to “find” a receptor. These features are highlighted in Figures 7 and 8, respectively. Figure 7 shows the increase of local receptor densities $\phi(\mathbf{r})/\phi_0$ (cf. eq 14) and Figure 8 shows the

probability distribution $P(\mathbf{r})$ for the chain end on the receptor surface as given in eq 28. We choose tether length $N = 50$, surface separation $h = 10$, receptor density $\phi_0 = 0.001$, and contrast the behavior between the annealed and quenched systems for three values of the binding energy ($\epsilon = 10kT$, $8kT$, and $5kT$). The distributions are calculated for lattice sites on the receptor surface along the y axis at $x = 0$. In a continuum model, because of spatial isotropy, this distribution would simply be the radial distribution. The lattice model introduces anisotropy. However, the qualitative behavior of the distribution along any other directions is similar and for simplicity, we still label the distance as the radial distance. (Alternatively we could perform a circular average, which would yield very similar results.)

We see that in the annealed case, the receptor density near the ligand is significantly higher than the average, implying that a receptor is more likely to be present due to the attractive binding by the ligand. For $\epsilon = 10kT$, the increase in probability is almost 10 fold at $\mathbf{r} = 0$. From eq 14 one sees that the total increase

$$\sum_{\mathbf{r}} (\phi(\mathbf{r}) - \phi_0) = \frac{\phi_0(1 - \phi_0)(e^{\beta\epsilon} - 1)Q_1/Q_0}{1 + \phi_0(e^{\beta\epsilon} - 1)Q_1/Q_0} \\ \lesssim f(1 - \phi_0),$$

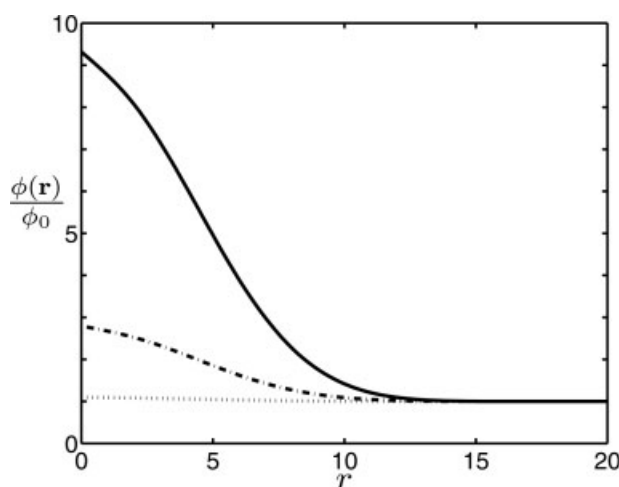


Figure 7. Local receptor density in the annealed case for three binding strengths $\epsilon = 10kT$ (solid), $\epsilon = 8kT$ (dashed), and $\epsilon = 5kT$ (dotted). $\phi(\mathbf{r})/\phi_0$ is plotted against the radial distance $\mathbf{r}$ from $(x = 0, y = 0, z = h)$ on the receptor surface. The other parameters are set at $N = 50$, $\phi_0 = 0.001$, and $h = 10$.

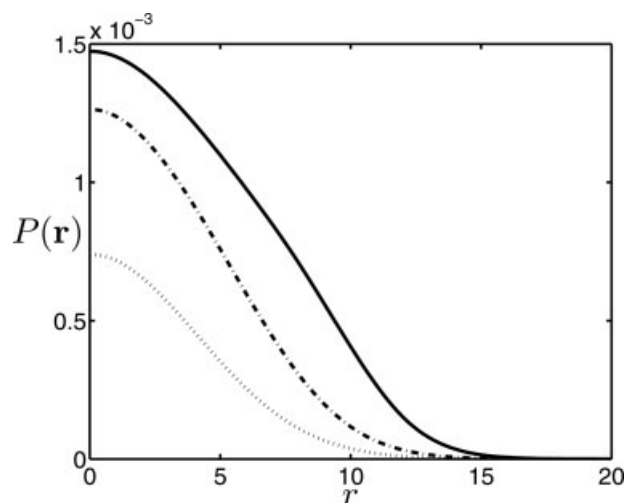


Figure 8. The probability distribution of the ligand reaching the receptor surface at radial distance $\mathbf{r}$ in the quenched case for the same parameters as in Figure 7. The curves are calculated using direct sampling.

where f is the binding fraction. This result states that in the strong binding regime ($\beta\epsilon \gg 1$), the total number of receptors near the ligand is increased by the maximum increase at each lattice site $(1 - \phi_0)$ multiplied by the binding fraction. Thus, the increase in the receptor density is essentially due to one receptor being attracted by the ligand to its vicinity. At small ϕ_0 , this increase is substantial; as ϕ_0 approaches 1, the effect becomes less appreciable.

In the quenched case, because the receptors are fixed, binding occurs by the tether polymer extending itself. The stronger the binding energy, the further the polymer stretches; this is evident by the significant increase in the half height width of the distribution as the binding energy increases. These features demonstrate the interesting interplay between the chain-conformation degrees of freedom and the microscopic ligand–receptor binding, which is manifested differently for the quenched and the annealed case.

CONCLUSIONS

In this article we have provided a simple model to illustrate how the physical flexibility of a polymer tether affects the effective range for ligand–receptor binding between two surfaces. This model allows analytical treatment, and captures some key features of the polymer-tethered

ligand–receptor binding that would be obscured in more detailed molecular models.

We have shown that the combination of conformation flexibility of the polymer chain and the short-ranged specific binding leads to nontrivial interaction profiles between the two surfaces. The tethered ligand forms a bridge with a finite probability once the functionalized surfaces come close enough so that the ligand–receptor binding energy gain can compensate for the entropy loss associated with chain stretching. For short to moderate chain lengths and large binding energies, binding can occur at distances considerably larger than the natural end-to-end distance of the polymer. At closer separations, the confinement of polymer gives rise to a strong repulsion. These two competing effects result in a free energy minimum at an equilibrium separation. From the interaction free energy profile, the critical force to separate the surfaces can be obtained. The use of a flexible tether turns the on-or-off type interaction in the molecular-level ligand–receptor bond into a softer, more gradual one and results in a critical force that is much weaker than that required to directly break the original ligand–receptor bond. These features are in qualitative agreement with experimental findings¹² and computer simulations results.³⁰

A special focus of this work is the difference between mobile (annealed) and immobile (quenched) receptors. In the annealed case, the mobility of the receptors allows for more effective binding, since a receptor can move to the location of the tethered ligand on the surface. Hence, by increasing the binding energy or bringing two surfaces closer, the binding fraction can reach unity. On the other hand, in the quenched case, even with the flexibility afforded by the polymer tether that allows the ligand to stretch to reach distant receptors, the binding fraction is limited by the availability of receptors within the reach of the tether. For sufficiently low receptor densities, the binding fraction will saturate at values less than unity regardless of how large the binding energy is or how close the two surfaces are. This qualitative difference between the two cases leads to large differences in the binding fraction and free energy profiles for low to moderate receptor densities. Aside from the obvious conclusion that mobile receptors result in more effective binding for the same set of parameters, we find that in the quenched case, for a given receptor density and binding energy, there exists an optimal chain length that gives the

lowest free energy minimum in the interaction free energy.

Another important result of this work is the density expansion method we have developed for treating the case of immobile receptors. Theoretical treatment of systems with quenched disorder usually requires the replica technique.³⁴ In particular, Chakraborty and coworkers used the replica technique to examine the issues of statistical pattern recognition between a random heteropolymer and a randomly patterned surface.³⁵ The density expansion provides an alternative route for treating quenched disorder, and is indeed more convenient when the randomness is due to the distribution of discrete particles. Even a two-term low density expansion turns out to work remarkably well when the binding fraction is not too high (<0.4).

Although explicit calculations in this work have made use of the ideal random walk model for the polymer, the theoretical formulation is independent of this assumption. Indeed, theoretical analysis in “Model and theory” shows that the only inputs from the polymer statistics are the partition function of a polymer confined between two surfaces with one end grafted on one surface and the probability distribution for the position of the free end. Both properties are independent of the binding aspect and can be obtained from independent sampling and enumeration. When a different polymer model is used, one only needs to substitute in the appropriate expression for the particular model. Such a strategy has been used successfully by Szeleifer and coworkers in the study of grafted polymer systems,³⁶ where self-avoiding polymer configurations are generated by enumeration and used in performing various averages for treating the interchain excluded volume effects in polymer brushes. Thus, excluded volume and chain flexibility effects can be easily accounted for. The formulation is also easily extended to continuum chain models, such as the freely jointed chain and the wormlike chain models.

We close with a brief comment on the connection between our single-ligand calculation and adhesion between two surfaces involving multiple ligands and receptors. As long as the receptors greatly outnumber the ligands, different ligand–receptor pairs can be considered independent. Several of the experimental systems^{11,12,25} appear to be in this regime. In these cases, the total interaction between the two surfaces is simply the single-ligand interaction

calculated here multiplied by the number of ligands. Indeed, the interpretation of the single-ligand result is probably best cast in this light (i.e., as an ensemble average of many independent ligands). When the number of ligands and receptors become comparable, different ligands can compete to bind with the same receptor and vice versa. This competition introduces nontrivial correlation effects. In the case where ligands and receptors are both immobile, it is conceivable that competitive binding can result in less-than-optimal ligand–receptor pairing patterns, that is, long-lived metastable adhesion states. Long-lived metastable states have been proposed in macromolecular recognition³⁷ and statistical pattern recognition between a heteropolymer and a randomly patterned surface,³⁸ where similar physics is at play. We will examine the multiple-ligand problem in a forthcoming paper.

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APPENDIX A: CALCULATION OF PARTITION FUNCTIONS Q_0 AND Q_1 BASED ON THE RANDOM WALK MODEL

In “Model and theory” we introduced the model for tethered binding based on the random walk model for the tether chain. Here, we give the analytical calculation of the partition functions in this model. Our calculation is based on the pioneering work of DiMarzio and Rubin.³⁹

We want to calculate the partition function Q_0 of an N -step random-walk polymer confined between two surfaces at $z = 0$ and $z = h$, with one end fixed at $\mathbf{r}_0 = (0, 0, 1)$. The boundary surfaces are assumed to be hard wall (impenetrable).

For a random walk polymer on a simple cubic lattice, the Green function $G(\mathbf{r}_1, \mathbf{r}_0; N)$ of N -step random walks starting at $\mathbf{r}_0$ and ending at $\mathbf{r}_1$ is given by

$$G(\mathbf{r}_1, \mathbf{r}_0; N) = \sum_{n_x+n_y+n_z=N} \frac{N!}{n_x!n_y!n_z!} \times G_x(x_1, x_0; n_x) G_y(y_1, y_0; n_y) G_z(z_1, z_0; n_z), \quad (\text{A1})$$

where $n_i \geq 0$ is the number of steps in the i th spatial direction, and G_i is the total number of

n_i -step random walks with given starting and ending positions. For convenience we slightly change our notations in this appendix: here $\mathbf{r} = (x, y, z)$ is the spatial vector instead of the planar vector on the surface $z = h$. In x and y directions, G_i is straightforwardly calculated by defining the numbers of “forward” and “backward” steps

$$n_x^f = \frac{n_x + x_1 - x_0}{2}, \quad n_x^b = \frac{n_x - x_1 + x_0}{2};$$

$$n_y^f = \frac{n_y + y_1 - y_0}{2}, \quad n_y^b = \frac{n_y - y_1 + y_0}{2}.$$

And $G_i (i = x, y)$ are given by

$$G_i(i_1, i_0; n_i) = \frac{n_i!}{n_i^f!n_i^b!}. \quad (\text{A2})$$

Because of the spatial constraints, the number of random walks in the z direction depends on the boundary conditions at the surfaces ($z = 0$ and $z = h$). For the reflecting boundary condition, $G_z(h, h - 1; 1) = G_z(h - 1, h; 1)$, $G_z(0, 1; 1) = G_z(1, 0; 1)$, G_z can be calculated using eq A2 exactly. Our hard wall boundary conditions correspond to $G_z(0, z_0; n_z) = G_z(h, z_0; n_z) = 0$ for any n_z or z_0 .⁴³ This problem can be solved exactly by recursion as follows.

First by symmetry between z_0 and z_1 , we can expand $G_z(z_1, z_0; n_z)$ into a double Fourier series

$$G_z(z_1, z_0; n_z) = \sum_{k=1}^{h-1} \sum_{m=1}^{h-1} A_{km}(n_z) \sin \frac{k\pi z_1}{h} \sin \frac{m\pi z_0}{h}. \quad (\text{A3})$$

The summation is over $1 \leq k \leq h - 1$ as $G(z_1, z_0; n_z)$ is nonvanishing only when $1 \leq z_0, z_1 \leq h - 1$.

Now applying the recursion relation for an ideal random walk in z direction

$$G_z(z_1, z_0; n_z) = G_z(z_1 - 1, z_0; n_z - 1) + G_z(z_1 + 1, z_0; n_z - 1), \quad (\text{A4})$$

we obtain

$$\begin{aligned}
 G_z(z_1, z_0; n_z) &= \sum_{k=1}^{h-1} \sum_{m=1}^{h-1} A_{km} (n_z - 1) \sin \frac{m\pi z_0}{h} \left[\sin \frac{k\pi (z_1 + 1)}{h} + \sin \frac{k\pi (z_1 - 1)}{h} \right] \\
 &= \sum_{k=1}^{h-1} \sum_{m=1}^{h-1} 2 \cos \frac{k\pi}{h} A_{km} (n_z - 1) \sin \frac{k\pi z_1}{h} \sin \frac{m\pi z_0}{h} \\
 &= \sum_{k=1}^{h-1} \sum_{m=1}^{h-1} A_{km} (0) \left(2 \cos \frac{k\pi}{h} \right)^{n_z} \sin \frac{k\pi z_1}{h} \sin \frac{m\pi z_0}{h}. \quad (\text{A5})
 \end{aligned}$$

Since G_z is a symmetric function of z_0 and z_1 , only those terms with $k = m$ should appear in eq A5, and we have

$$\begin{aligned}
 G_z(z_1, z_0; n_z) &= \sum_{k=1}^{h-1} A_k \left(2 \cos \frac{k\pi}{h} \right)^{n_z} \\
 &\quad \times \sin \frac{k\pi z_1}{h} \sin \frac{k\pi z_0}{h}, \quad (\text{A6})
 \end{aligned}$$

where

$$A_k = A_{km}(0)$$

can be fixed by imposing the initial condition that

$$G_z(z_0, z_0; 0) = 1,$$

which gives

$$\begin{aligned}
 1 &= \sum_{k=1}^{h-1} A_k \sin^2 \frac{k\pi z_0}{h} = \sum_{k=1}^{h-1} A_k \frac{1 - \cos \frac{2k\pi z_0}{h}}{2}, \\
 z_0 &= 1, 2, \dots, h-1. \quad (\text{A7})
 \end{aligned}$$

Inspecting this equation and noticing that

$$\sum_{k=1}^{h-1} \cos \frac{2kz_0\pi}{h} = -1,$$

one immediately finds a solution

$$A_k = 2/h.$$

Because of the linear independence of the Fourier series, this solution is also unique. Hence,

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we obtain G_z to be

$$G_z(z_1, z_0; n_z) = \frac{2}{h} \sum_{k=1}^{h-1} \sin \frac{k\pi z_1}{h} \left(2 \cos \frac{k\pi}{h} \right)^{n_z} \sin \frac{k\pi z_0}{h}. \quad (\text{A8})$$

The partition function Q_0 can be calculated from eqs A1 and A8 straightforwardly as a summation over the end positions of the polymer.

APPENDIX B: LOW DENSITY EXPANSION

In this appendix we give the details of the low density expansion in “Immobile receptors”. To calculate the virial coefficients B_n defined in eq 20, we need to evaluate the quenched average in eq 19

$$\begin{aligned}
 &\left\langle \ln \left[1 + (e^{\beta\epsilon} - 1) \sum_{\mathbf{r}} \sigma(\mathbf{r}) P_0(\mathbf{r}) \right] \right\rangle_{\{\sigma\}} \\
 &= \sum_{0 \leq n \leq M} \sum_{k=0}^{M-n} \frac{(-1)^k (M-n)!}{n! k! (M-n-k)!} \phi_0^{n+k} \\
 &\quad \sum_{\mathbf{r}_1} \sum_{\mathbf{r}_2 \neq \mathbf{r}_1} \dots \sum_{\mathbf{r}_n \neq \mathbf{r}_1, \dots, \mathbf{r}_{n-1}} \ln \left[1 + \sum_{i=1}^n P_0(\mathbf{r}_i) (e^{\beta\epsilon} - 1) \right].
 \end{aligned}$$

To the 1st order, there is only one term corresponding to $n = 1, k = 0$,

$$\phi_0 \sum_{\mathbf{r}} \ln [1 + P_0(\mathbf{r}) (e^{\beta\epsilon} - 1)]. \quad (\text{A9})$$

To the 2nd order, there are two contributions, one from $n = 1, k = 1$

$$-(M-1) \sum_{\mathbf{r}} \ln[1 + (e^{\beta\epsilon} - 1)P_0(\mathbf{r})],$$

and the other from $n = 2, k = 0$

$$\frac{1}{2} \sum_{\mathbf{r}_1} \sum_{\mathbf{r}_2 \neq \mathbf{r}_1} \ln\{1 + (e^{\beta\epsilon} - 1)[P_0(\mathbf{r}_1) + P_0(\mathbf{r}_2)]\}.$$

One can collect the terms and rearrange them to give

$$\begin{aligned} f &= -\frac{\partial F}{\partial \epsilon} \\ &= \phi_0 \sum_{\mathbf{r}} \frac{P_0(\mathbf{r})e^{\beta\epsilon}}{1 + P_0(\mathbf{r})(e^{\beta\epsilon} - 1)} + \frac{\phi_0^2}{2} \left\{ \sum_{\mathbf{r}} \left[\frac{2P_0(\mathbf{r})e^{\beta\epsilon}}{1 + P_0(\mathbf{r})(e^{\beta\epsilon} - 1)} - \frac{2P_0(\mathbf{r})e^{\beta\epsilon}}{1 + 2P_0(\mathbf{r})(e^{\beta\epsilon} - 1)} \right] \right. \\ &\quad \left. + \sum_{\mathbf{r}_1} \sum_{\mathbf{r}_2} \left[\frac{[P_0(\mathbf{r}_1) + P_0(\mathbf{r}_2)]e^{\beta\epsilon}}{1 + [P_0(\mathbf{r}_1) + P_0(\mathbf{r}_2)](e^{\beta\epsilon} - 1)} - \frac{P_0(\mathbf{r}_1)e^{\beta\epsilon}}{1 + P_0(\mathbf{r}_1)(e^{\beta\epsilon} - 1)} - \frac{P_0(\mathbf{r}_2)e^{\beta\epsilon}}{1 + P_0(\mathbf{r}_2)(e^{\beta\epsilon} - 1)} \right] \right\} + O(\phi_0^3). \end{aligned} \quad (\text{A10})$$

In general, one can work out the higher order terms in the expansion recursively. Specifically the summation

$$\sum_{\mathbf{r}_1} \sum_{\mathbf{r}_2 \neq \mathbf{r}_1} \cdots \sum_{\mathbf{r}_n \neq \mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_{n-1}}$$

can be expressed using the sums

$$\sum_{\mathbf{r}_1}, \sum_{\mathbf{r}_1, \mathbf{r}_2}, \cdots, \sum_{\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_n}$$

through recursion relations

$$\sum_n' = \sum_{\mathbf{r}_1} \sum_{\mathbf{r}_2 \neq \mathbf{r}_1} \cdots \sum_{\mathbf{r}_n \neq \mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_{n-1}} \quad (\text{A11})$$

$$\begin{aligned} &= \sum_{\mathbf{r}_1} \sum_{n-1}' - (n-1) \sum_{\mathbf{r}_1 = \mathbf{r}_2} \sum_{n-2}' + \cdots \\ &\quad + (-1)^{n-1} (n-1)! \sum_{\mathbf{r}_1 = \mathbf{r}_2 = \cdots = \mathbf{r}_n}. \end{aligned} \quad (\text{A12})$$

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$$\begin{aligned} & -\frac{\phi_0^2}{2} \left\{ \sum_{\mathbf{r}_1} \sum_{\mathbf{r}_2} \ln \frac{[1 + P_0(\mathbf{r}_1)(e^{\beta\epsilon} - 1)][1 + P_0(\mathbf{r}_2)(e^{\beta\epsilon} - 1)]}{1 + [P_0(\mathbf{r}_1) + P_0(\mathbf{r}_2)](e^{\beta\epsilon} - 1)} \right. \\ & \quad \left. - \sum_{\mathbf{r}} \ln \frac{[1 + P_0(\mathbf{r})(e^{\beta\epsilon} - 1)]^2}{1 + 2P_0(\mathbf{r})(e^{\beta\epsilon} - 1)} \right\}. \end{aligned}$$

Similarly, one can obtain the binding fraction to ϕ_0^2 :

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40. Strictly speaking, the results for these two cases are identical only if the number of receptors and the number of ligands are fixed. The results will be different if any one of the species is connected to a reservoir. This point will be discussed further in a forthcoming paper.
41. Molecular ligand–receptor binding force is not well defined in the lattice model. However, an estimate can be made by taking the binding energy divided by the lattice spacing. For a binding energy of $\epsilon = 15kT$ with $T \approx 300$ K, taking $a = 0.5$ nm, we obtain a force of ~ 120 pN.
42. Inspection of eqs 15 and 16 reveals that the average can be calculated in terms of the single random variable $\eta = \sum_{\mathbf{r}} \sigma(\mathbf{r})P_0(\mathbf{r})$. It can be shown that the variance of η is inversely proportional to the average number of receptors within the reach of the ligand. Therefore, for high receptor density or long tether length, the fluctuation of η around its mean value is small, and η in the quenched average can be replaced by its mean value, which results in identical expressions to the annealed case (eqs 11 and 13). Although not shown here, we anticipate based on similar reasoning, that increasing the chain length will have similar effects, and for sufficiently long chains, “quenched average” becomes the same as “annealed average”.
43. Note that there is a slight difference here as we have put the hard wall boundaries at $z = 0$ and $z = h$ for convenience in calculating the Green function of the confined polymer. In the main text, the polymer segment can reach $z = 0$ and $z = h$, corresponding to hard walls placed at $z = -1$ and $z = h + 1$ (cf. eq 2 in “Model description”). Therefore, the h here equals to $h + 2$ in the main text.